

PhD Topology Exam

August 2025

Be neat, do not write too small, and leave margins.

For the first five problems, show all of your work and support all statements. Use a separate sheet of paper for each problem. (Each problem 10 points.)

1. Let (X, d) be a compact metric space. Let $f: X \rightarrow X$ and suppose there is some $c < 1$ with $d(f(x), f(y)) \leq c \cdot d(x, y)$ for all $x, y \in X$. Prove there exists a unique $x \in X$ with $f(x) = x$.
2. Pick a CW complex structure on the Klein bottle K , and then use cellular homology to compute $H_i(K; \mathbb{Z})$ for $i = 0, 1, 2$.
3. (a) Show that the antipodal map $-1: S^n \rightarrow S^n$ defined by $-1(x) = -x$ has degree $(-1)^{n+1}$.
(b) Prove that S^n has a continuous nonzero tangent vector field if and only if n is odd.
4. Let M be a compact orientable n -manifold without boundary. Prove that if n is odd, then the Euler characteristic $\chi(M)$ is zero.
5. Let M_1 and M_2 be closed connected n -manifolds, and let $M_1 \# M_2$ be their connected sum obtained by removing a ball from each and gluing along the resulting boundary spheres. Show that there are isomorphisms $H_i(M_1 \# M_2; \mathbb{Z}) \cong H_i(M_1; \mathbb{Z}) \oplus H_i(M_2; \mathbb{Z})$ for $0 < i < n$, with one exception: If both M_1 and M_2 are nonorientable, then $H_{n-1}(M_1 \# M_2; \mathbb{Z})$ is obtained from $H_{n-1}(M_1; \mathbb{Z}) \oplus H_{n-1}(M_2; \mathbb{Z})$ by replacing one of the two $\mathbb{Z}_2$ summands by a $\mathbb{Z}$ summand.

The remaining problems are on the following page.

Answer the following ten problems with complete definitions, complete statements, an example, or a short proof. (Each problem 5 points.)

6. Show that if $g: S^2 \rightarrow S^2$ is continuous and $g(x) \neq g(-x)$ for all $x \in S^2$, then g is surjective.
7. State the Tychonoff Theorem.
8. Show that $\mathbb{R}^\omega$ with the box topology is not connected.
9. Let N_g be the nonorientable surface of genus g , i.e., the g -fold connected sum of projective planes. Use the Seifert van-Kampen theorem to derive the fundamental group of N_g for $g \geq 1$.
10. Let X be two unlinked circles in $\mathbb{R}^3$ and let Y be two simply linked circles. Show that the complements $\mathbb{R}^3 \setminus X$ and $\mathbb{R}^3 \setminus Y$ are not homotopy equivalent.
11. Show that if n is even, then every map $f: \mathbb{RP}^n \rightarrow \mathbb{RP}^n$ has a fixed point.
12. Let $p_1: (\tilde{X}_1, \tilde{x}_1) \rightarrow (X, x_0)$ and $p_2: (\tilde{X}_2, \tilde{x}_2) \rightarrow (X, x_0)$ be covering space maps, where all of the involved spaces are path-connected and locally path-connected. An *isomorphism of covering spaces* $f: (\tilde{X}_1, \tilde{x}_1) \rightarrow (\tilde{X}_2, \tilde{x}_2)$ is a homeomorphism satisfying $p_1 = p_2 f$. Prove the covering spaces p_1 and p_2 are isomorphic if and only if $p_{1*}(\pi_1(\tilde{X}_1, \tilde{x}_1)) = p_{2*}(\pi_1(\tilde{X}_2, \tilde{x}_2))$.
13. Define what it means for a subspace $A \subseteq X$ to be a *retract*. Give an example of a retract $A \subseteq X$ that is not a deformation retract.
14. Use the Tietze extension theorem to prove Urysohn's lemma.
15. What are the cohomology groups and cup product structure in $H^*((S^2 \times S^8) \# (S^4 \times S^6); \mathbb{Z})$, where $\#$ denotes the connected sum of the manifolds? You do not need to prove your answer.